

Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?
Answer Choices:
A. 2.48×10^{-8} cm
B. 5.92×10^7 cm/sec
C. 8.66×10^{-6} cm
D. 3.14×10^{-5} cm
E. 9.81×10^{-4} cm
F. 5.27×10^{-9} m
G. 1.23×10^{-7} cm
H. 4.14×10^{-10} m
I. 6.625×10^{-27} erg sec
J. 0.512×10^6 eV

Reasoning + Answer:

</tool_call> reasoning process here
The de Broglie wavelength of a particle is given by the formula :
 $\lambda = h / p$

where :
- h is Planck's constant = 6.626×10^{-34} J·s (or 6.626×10^{-27} erg·s),
- p is the momentum of the particle .

For a non-relativistic electron , kinetic energy $K = (1/2)mv^2$, and momentum $p = mv$.

So , we can write :
 $p = \sqrt{2mK}$

Therefore , de Broglie wavelength :

$$\lambda = h / \sqrt{2mK}$$

First , convert kinetic energy to joules :
 $1\text{ eV} = 1.602 \times 10^{-19}\text{ J}$

$$\text{So } K = 1.602 \times 10^{-19}\text{ J}$$

$$\text{Mass of electron } m = 9.109 \times 10^{-31}\text{ kg}$$

Now plug into the formula :

$$\lambda = (6.626 \times 10^{-34}) / \sqrt{2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19}}$$

Compute inside square root :
 $2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19} = 2.92 \times 10^{-49}$

$$\sqrt{2.92 \times 10^{-49}} \approx 5.40 \times 10^{-25}$$

Then :
 $\lambda = 6.626 \times 10^{-34} / 5.40 \times 10^{-25} \approx 1.227 \times 10^{-9}\text{ m}$

$$\text{Convert to cm: } 1.227 \times 10^{-9}\text{ m} = 1.227 \times 10^{-7}\text{ cm}$$

Now , check options in cm :

A. 2.48×10^{-8} cm $\approx 2.48 \times 10^{-8}$
B. 5.92×10^7 cm $\approx$ too large
C. 8.66×10^{-6} cm
D. 3.14×10^{-5} cm
E. 9.81×10^{-4} cm
F. 5.27×10^{-9} m $\approx 5.27 \times 10^{-10}$ m = 5.27×10^{-8} cm $\approx$ not matching
G. 1.23×10^{-7} cm $\approx$ this is 1.23×10^{-7} cm $\approx$ which is 1.23×10^{-6} m $\approx 1.23 \times 10^{-9}$ m
Wait $\hat{G}$ let's verify conversion :

$1.227 \times 10^{-9}\text{ m} = 1.227 \times 10^{-7}\text{ cm}$?
Yes because $1\text{ m} = 100\text{ cm}$ $\approx 1 \times 10^{-9}\text{ m} = 1 \times 10^{-7}\text{ cm}$ $\approx$ so $1.227 \times 10^{-9}\text{ m} = 1.227 \times 10^{-7}\text{ cm}$

$$\text{So } \lambda \approx 1.23 \times 10^{-7}\text{ cm}$$

Option G is : $1.23 \times 10^{-7}\text{ cm}$ $\approx$ matches

$$\text{Convert option G to m: } 1.23 \times 10^{-7}\text{ cm} = 1.23 \times 10^{-9}\text{ m} \approx \text{known as } 1.23 \times 10^{-9}\text{ m?}$$

$$\text{Wait } 10^{-7}\text{ cm} = 10^{-9}\text{ m} \approx \text{so } 1.23 \times 10^{-7}\text{ cm} = 1.23 \times 10^{-9}\text{ m} \approx \text{erg , yes}$$

$$\text{Option H is } 4.14 \times 10^{-10}\text{ m} = 4.14 \times 10^{-8}\text{ cm} \approx \text{not same}$$

$$\text{So wavelength} = \sim 1.23 \times 10^{-9}\text{ m}$$